# Classical Forecasting vs. Time-Series Foundation Models

Comparing established methods and pretrained models across 12 datasets

**5 classical methods** | **3 foundation-model families** | **12 datasets**

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# Why this comparison matters

**Do time-series foundation models consistently outperform classical forecasting approaches across datasets with different statistical characteristics?**

**Zero-shot promise**

Can a pretrained model transfer across domains, frequencies, and horizons without dataset-specific training?

**Model families**

Compare that promise against established statistical, algorithmic, and neural forecasting approaches.

**Different metrics**

Separate point accuracy from quantile and predictive-distribution quality; they answer different questions.

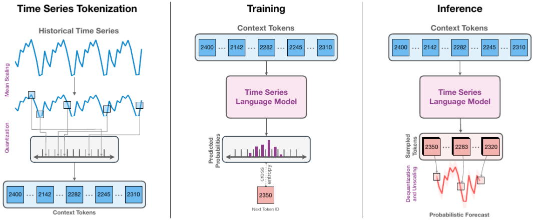
**Statistical tests**

Test whether observed performance differences are systematic or compatible with sampling variation.

# Classical and conventional forecasting models

| Model | Forecasting mechanism | Main strength | Main limitation |
|---|---|---|---|
| Seasonal Naive | Repeats the most recent seasonal cycle. | Transparent and robust reference for seasonal data. | Cannot adapt to trend or changing seasonal structure. |
| ARIMA $(2, 1, 2)$ | Models differenced values through autoregressive and moving-average terms. | Captures short-range linear dynamics with a compact local model. | Fixed order; limited under nonlinear behavior or regime changes. |
| AutoARIMA | Searches candidate ARIMA orders and selects the best fit by AIC. | Automates model specification for each series. | Search can be costly or unstable on short and noisy histories. |
| Prophet | Decomposes a series into trend, changepoints, and calendar seasonalities. | Interpretable components and flexible trend changes. | A rigid additive decomposition may miss local dynamics. |
| DeepAR | Learns an autoregressive recurrent network under a probabilistic likelihood. | Represents nonlinear temporal relationships and uncertainty. | Requires training and hyperparameter choices for each task. |

# Chronos: forecasting as a language-model task



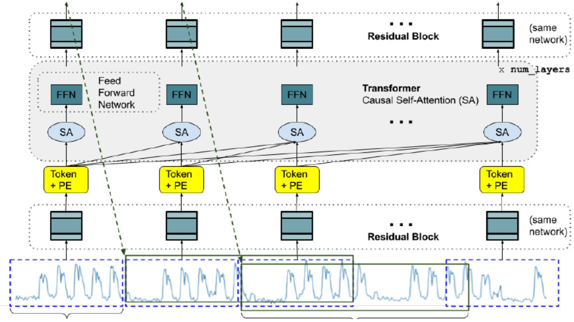

- **Representation:** mean-scale the series, quantize values into bins, and map them to a fixed token vocabulary.
- **Architecture:** a T5 encoder–decoder learns next-token probabilities using cross-entropy.
- **Forecast:** autoregressive token samples are dequantized into trajectories. Inference is **stochastic**; the sampled trajectories define a **probabilistic forecast**.

## Chronos–T5 variants

| Variant | Params | Enc./Dec. |
|---|---|---|
| Tiny | 8M | 4 / 4 |
| Mini | 20M | 4 / 4 |
| Small | 46M | 6 / 6 |
| Base | 200M | 12 / 12 |
| Large | 710M | 24 / 24 |

The variants use the same tokenization and forecasting mechanism. Capacity increases through greater depth and width.

# TimesFM: patch the signal, decode the future




## Patch tokens

Non-overlapping blocks of 32 observations are normalized and projected into token embeddings.

## Decoder-only

Causal self-attention lets each patch representation use only its available history.

## Multi-horizon

Each output head predicts a block of future time points; blocks are composed to reach the requested horizon.

**Forecast behavior**

TimesFM 2.5 directly outputs the mean and quantiles q10–q90. The values are **probabilistic forecasts**, but inference is **deterministic**: the same input returns the same distribution.

# Moirai: one model for any frequency and variate count


## Multiple patch sizes

A bank of input/output projections supports several patch lengths. The model can match its temporal resolution to series with different sampling frequencies and local dynamics.

## Any-variate attention

Patches from every variable are flattened into one token sequence. Time and variable identifiers preserve where each patch came from, so attention can learn both within- and cross-variable dependence.

## Distribution forecast

Masked future patches are decoded into an uncertainty representation. The original Moirai uses a flexible mixture density; Moirai 2.0 directly predicts quantiles. Inference is deterministic.

# Twelve datasets I: competition and archive panels

| Dataset | Domain | Full dataset | Series used | Freq. / H | Observations | Statistical characteristics |
|---|---|---|---|---|---|---|
| M1 Yearly | Business / economic | 181 univariate series | 20 | Y / 6 | 15–58 | Trends, level shifts, weak or absent seasonality |
| M3 Quarterly | Business / economic | 756 univariate series | 20 | Q / 8 | 24–72 points | Quarterly cycles, trend, nonstationarity |
| M4 Hourly | Mixed competition | 414 univariate series | 20 | h / 48 | 748–1,008 points | High frequency, daily seasonality, noise |
| Weather (rain) | Environment | Rain subset of 3,010 series | 20 | D / 30 | 1,332–65,981 | Zero inflation, intermittency, seasonal weather |
| Car Parts | Retail / inventory | 2,674 univariate series | 20 | M / 12 | 51 each | Sparse intermittent demand, many zeros |
| NN5 Weekly | Banking / ATM | 111 univariate series | 20 | W / 8 | about 113 | Annual seasonality, holidays, demand volatility |

# Twelve datasets II: health, infrastructure, and macroeconomics

| Dataset | Domain | Full dataset | Series used | Freq. / H | Observations | Statistical characteristics |
|---|---|---|---|---|---|---|
| National Illness | Public health | 1 series, 7 variables | 1 target | W / 24 | 966 weeks | Epidemic peaks, annual cycle, regime change |
| Traffic | Transport | 1 series, 862 sensors | 5 targets | h / 24 | 17,544 hours | Strong daily/weekly cycles, correlated sensors |
| PSM | Server telemetry | 1 series, 25 variables | 5 targets | h / 24 | 132,481 min. rows, resampled h | Anomalies, abrupt shifts, high-frequency noise |
| Elexon Demand | Energy | 1 national demand series | 1 | D / 30 | 2024–2026 (730 days) | Weekly cycle, weather/calendar effects, nonstationarity |
| USGS Streamflow | Hydrology | 5 gauge series | 5 | D / 30 | 2015–2026 ($\sim$4,230 days) | Annual hydrology, skew, floods/droughts, gaps |
| BLS Macro | Macro | 3 indicator series | 3 | M / 6 | 2010–2026 ($\sim$199 months) | Trends, structural breaks, heterogeneous scales |

# Point and probabilistic forecast evaluation

## MASE point

$$\mathrm{MASE} = \frac{H^{-1}\sum_{h=1}^{H}|y_h - \hat{y}_h|}{(T-m)^{-1}\sum_{t=m+1}^{T}|y_t - y_{t-m}|}$$

Error scaled by an in-sample seasonal-naive error. It is comparable across units and datasets.

$< 1$ beats the scale; $= 1$ matches it; $> 1$ is worse.

## WQL quantiles

$$\mathrm{WQL} = \frac{1}{|Q|}\sum_{q\in Q}\frac{2\sum_h \rho_q(y_h - \hat{y}_{q,h})}{\sum_h |y_h|}$$

Weighted quantile (pinball) loss on $Q = \{0.1, \ldots, 0.9\}$, normalized by target magnitude.

Multiple quantiles reveal asymmetric risk and interval quality.

## CRPS distribution

$$\mathrm{CRPS}(F, y) = \int_{-\infty}^{\infty}\left(F(z) - \mathbf{1}\{y \le z\}\right)^2 dz$$

Scores the entire predictive CDF against the observation; rewards **calibration** and **sharpness** jointly.

Implemented as a finite-quantile approximation from q10–q90.

**Central forecast**
How accurate is the median path?

**Quantile accuracy**
Where are under- and over-predictions?

**Distribution quality**
How well does the predictive CDF match reality?

**Together they separate scale-free point error from quantile and distributional quality; lower is better.**

# Statistical inference for forecast comparisons

## Diebold–Mariano

### Which of two models is better?

$$d_t = L_t(A) - L_t(B), \qquad H_0 : \mathbb{E}[d_t] = 0$$

**Use:** compare A and B on the same forecast origins. The variance estimate accounts for serial dependence in $d_t$.

**Decision**
$p < \alpha$ and $\bar{d} < 0$: A has significantly lower expected loss.

## SPA

### Does any alternative beat a benchmark?

$$d_{k,t} = L_t(B) - L_t(k)$$

$$H_0 : \max_k \mathbb{E}[d_{k,t}] \leq 0$$

**Use:** compare benchmark B with many challengers at once. The bootstrap corrects for searching over multiple models.

**Decision**
Reject $H_0$: at least one challenger has lower expected loss than B.

## Model Confidence Set

### Which models remain plausible best models?

$$\mathcal{M}_0 \supset \cdots \supset \widehat{\mathcal{M}}_{1-\alpha}$$

**Use:** begin with every model. Repeatedly test equal predictive ability and remove the worst-supported model.

**Output**
$\widehat{\mathcal{M}}_{1-\alpha}$ is a set of models that cannot be distinguished from the best at level $\alpha$.

# Empirical winners across the three core losses

| Dataset | MASE | WQL | CRPS |
|---|---|---|---|
| M1 Yearly | **TimesFM** | **ARIMA** | **ARIMA** |
| M4 Hourly | **Chronos Base** | **Chronos Base** | **Chronos Base** |
| Weather | **Chronos Base** | **Moirai** | **Moirai** |
| M3 Quarterly | **TimesFM** | **TimesFM** | **TimesFM** |
| Car Parts | **Moirai** | **Moirai** | **Moirai** |
| NN5 Weekly | **TimesFM** | **TimesFM** | **TimesFM** |
| National Illness | **TimesFM** | **TimesFM** | **TimesFM** |
| Traffic | **Moirai** | **Moirai** | **Moirai** |
| PSM | **Chronos Small** | **Chronos Small** | **Chronos Small** |
| Elexon Demand | **TimesFM** | **TimesFM** | **TimesFM** |
| USGS Streamflow | **Moirai** | **TimesFM** | **TimesFM** |
| BLS Macro | **Auto-ARIMA** | **TimesFM** | **TimesFM** |

**foundation model** **statistical model**

Each cell is the lowest aggregate loss among 11 models; this is a **descriptive ranking**, not a significance claim.

# How often each model wins a dataset–metric cell

| Model | MASE | WQL | CRPS | Total |
|---|---|---|---|---|
| TimesFM 2.5 200M | 5 | 6 | 6 | **17** |
| Moirai 2.0 Small | 3 | 3 | 3 | **9** |
| Chronos Base | 2 | 1 | 1 | **4** |
| Chronos Small | 1 | 1 | 1 | **3** |
| ARIMA(2,1,2) | 0 | 1 | 1 | **2** |
| Auto-ARIMA | 1 | 0 | 0 | **1** |
| Chronos Mini | 0 | 0 | 0 | **0** |
| Chronos Tiny | 0 | 0 | 0 | **0** |
| DeepAR | 0 | 0 | 0 | **0** |
| Prophet | 0 | 0 | 0 | **0** |
| Seasonal Naive | 0 | 0 | 0 | **0** |
| **Column total** | **12** | **12** | **12** | **36** |

## 36 winner cells

- 12 datasets $\times$ 3 core losses.
- A win is the lowest aggregate loss for one dataset and metric.
- TimesFM leads with **17/36**; Moirai follows with **9/36**.
- Foundation models occupy **33/36** empirical winner cells.
- Counts summarize breadth, not statistical significance; the losses are correlated and cells are not independent experiments.

# When every foundation model beats every comparator

**Strict family dominance:** $\max_{m\in \mathrm{FM}} L(m) < \min_{m\in \mathrm{C}} L(m)$ — even the weakest foundation model has lower aggregate loss than the strongest classical/conventional comparator.

| Dataset | MASE | WQL | CRPS |
|---|---|---|---|
| M4 Hourly | ✓ Chronos Base | ✓ Chronos Base | ✓ Chronos Base |
| PSM† | ✓ Chronos Small | ✓ Chronos Small | ✓ Chronos Small |
| Elexon Demand† | ✓ TimesFM | ✓ TimesFM | ✓ TimesFM |
| USGS Streamflow† | ✓ Moirai | ✓ TimesFM | ✓ TimesFM |

**12 cells**
Four datasets satisfy strict dominance on all three losses.

**+4 cells**
Car Parts and Traffic satisfy it for WQL and CRPS only.

† Descriptive low-power case; among the four complete-dominance datasets, only M4 Hourly has at least 20 paired tasks.

# Foundation model first, but the ranking is interleaved

The lowest loss belongs to a foundation model, yet the best conventional model still outranks one or more foundation models.

| Example | Foundation leader | Best conventional | Rank | FMs below it |
|---|---|---|---|---|
| M1 Yearly / MASE | TimesFM 3.129 | Prophet 3.184 | 2nd | 5 of 6 |
| M3 Quarterly / WQL | TimesFM 0.0776 | Auto-ARIMA 0.0850 | 2nd | 5 of 6 |
| NN5 Weekly / CRPS | TimesFM 11.855 | DeepAR 13.695 | 5th | 2 of 6 |
| Weather / CRPS | Moirai 1.188 | DeepAR 1.305 | 6th | 1 of 6 |

**17 of 36 cells** A foundation model ranks first, but strict family dominance fails.

**Interpretation:** a first-place foundation model does not imply that the foundation-model family has displaced established forecasting methods.

# Where a classical model beats every foundation model

**3 of 36 cells**

| Dataset | Loss | Classical winner | Best foundation model | Lower loss |
|---|---|---|---|---|
| M1 Yearly | WQL | ARIMA 0.0827 | Chronos Tiny 0.0944 | **12.4%** |
| M1 Yearly | CRPS | ARIMA 202,882 | Chronos Tiny 231,712 | **12.4%** |
| BLS Macro† | MASE | Auto-ARIMA 0.141 | TimesFM 0.174 | **18.8%** |

**M1 is the substantive exception**
ARIMA leads both probabilistic losses across 60 paired tasks.

**BLS remains tentative**
Its MASE result is based on only seven paired tasks.

Values are aggregate losses; percentages compare the classical winner with the best-performing foundation model. Lower is better.

# Is the empirical benchmark narrative enough?

## Are the observed winners meaningfully winning?

### Empirical narrative

- Lowest aggregate metric values
- Average ranks and winner counts
- Breadth across datasets and domains
- Point, quantile, and distributional losses

### Unanswered questions

- Is the observed gap distinguishable from sampling variation?
- Does the claim survive comparison with ten alternatives?
- Which models remain statistically plausible winners?
- Are small-sample results sufficiently powered?

**empirical winner** ⟶ **SPA + DM** ⟶ **95% MCS** ⟶ **defensible claim**

# MASE: statistical evidence behind the empirical winner

| Dataset | Empirical winner | SPA | DM* | MCS size |
|---|---|---|---|---|
| M1 Yearly | TimesFM | 3/10 | 1/10 | 9 |
| M4 Hourly | Chronos Base | 5/10 | 10/10 | 6 |
| Weather | Chronos Base | 4/10 | 6/10 | 8 |
| M3 Quarterly | TimesFM | 9/10 | 4/10 | 4 |
| Car Parts | Moirai | 4/10 | 4/10 | 7 |
| NN5 Weekly | **TimesFM** | **10/10** | **10/10** | **1** |
| National Illness† | TimesFM | 1/10 | — | 7 |
| Traffic† | Moirai | 5/10 | 3/10 | 6 |
| PSM† | Chronos Small | 4/10 | 4/10 | 7 |
| Elexon Demand† | TimesFM | 9/10 | — | 3 |
| USGS Streamflow† | Moirai | 3/10 | 1/10 | 8 |
| BLS Macro† | Auto-ARIMA | 5/10 | — | 5 |

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run ($< 8$ ordered tasks). †Low power ($< 20$ paired tasks).

**Strict separation occurs only on NN5 Weekly: TimesFM beats all ten alternatives by SPA and is the sole MCS member.**

# WQL: statistical evidence behind the empirical winner

| Dataset | Empirical winner | SPA | DM* | MCS size |
|---|---|---|---|---|
| M1 Yearly | ARIMA | 0/10 | 0/10 | 11 |
| M4 Hourly | Chronos Base | 0/10 | 0/10 | 11 |
| Weather | Moirai | 9/10 | 9/10 | 2 |
| M3 Quarterly | TimesFM | 9/10 | 7/10 | 2 |
| Car Parts | Moirai | 4/10 | 5/10 | 7 |
| NN5 Weekly | **TimesFM** | **10/10** | **10/10** | **1** |
| National Illness† | TimesFM | 1/10 | — | 5 |
| Traffic† | Moirai | 6/10 | 3/10 | 5 |
| PSM† | Chronos Small | 4/10 | 3/10 | 7 |
| Elexon Demand† | TimesFM | 9/10 | — | 2 |
| USGS Streamflow† | TimesFM | 0/10 | 0/10 | 11 |
| BLS Macro† | TimesFM | 0/10 | — | 11 |

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run ($< 8$ ordered tasks). †Low power ($< 20$ paired tasks).

**Strict separation occurs only on NN5 Weekly; Weather and M3 show strong but not unique leadership.**

# CRPS: statistical evidence behind the empirical winner

| Dataset | Empirical winner | SPA | DM* | MCS size |
|---|---|---|---|---|
| M1 Yearly | ARIMA | 0/10 | 0/10 | 11 |
| M4 Hourly | Chronos Base | 0/10 | 0/10 | 11 |
| Weather | **Moirai** | **10/10** | **10/10** | **1** |
| M3 Quarterly | TimesFM | 9/10 | 7/10 | 5 |
| Car Parts | Moirai | 4/10 | 10/10 | 7 |
| NN5 Weekly | **TimesFM** | **10/10** | **10/10** | **1** |
| National Illness† | TimesFM | 1/10 | — | 5 |
| Traffic† | Moirai | 6/10 | 3/10 | 5 |
| PSM† | Chronos Small | 4/10 | 3/10 | 7 |
| Elexon Demand† | TimesFM | 9/10 | — | 2 |
| USGS Streamflow† | TimesFM | 0/10 | 0/10 | 11 |
| BLS Macro† | TimesFM | 1/10 | — | 11 |

SPA/DM = Holm-adjusted significant wins out of 10; MCS = models retained in the 95% set. *DM is exploratory; — = not run ($< 8$ ordered tasks). †Low power ($< 20$ paired tasks).

**Weather (Moirai) and NN5 Weekly (TimesFM) are the two clean CRPS results: 10/10 SPA wins and singleton confidence sets.**

# Conclusion: leadership is broad, certainty is concentrated

**33/36**
foundation-model wins

**16/36**
family-dominance cells

**3/36**
classical-model wins

**4/36**
strictly separated wins

**Foundation models lead—but the advantage is dataset-, metric-, and evidence-dependent.**

### What the benchmark supports

- TimesFM has the broadest empirical coverage; Moirai and Chronos lead in distinct domains.
- Clean separation appears for all three NN5 losses and Weather CRPS.

### What limits the claim

- Classical models lead M1 WQL/CRPS and BLS MASE; most other winners share their MCS with alternatives.
- Low-power datasets and exploratory DM results cannot support universal superiority claims.